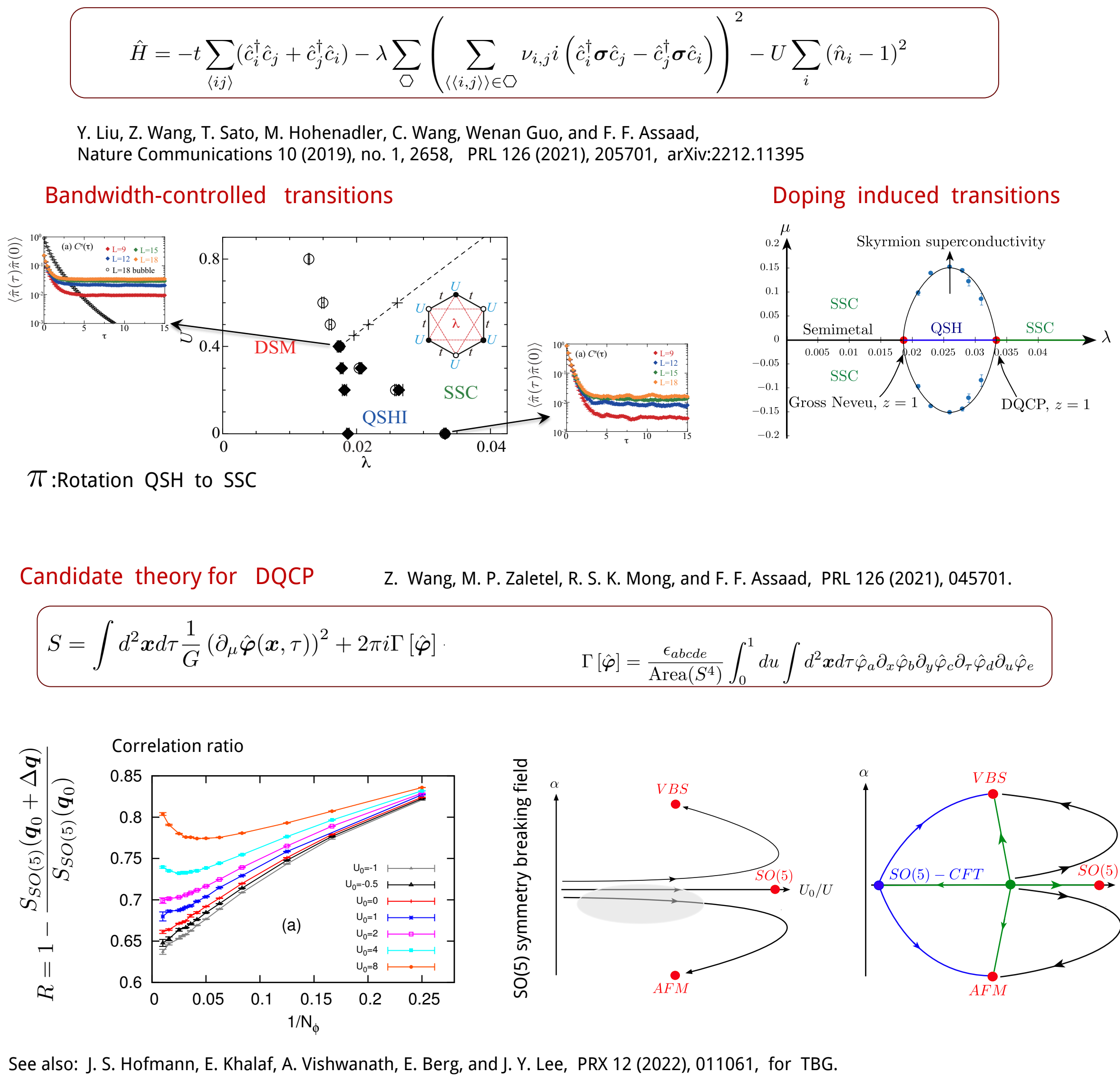


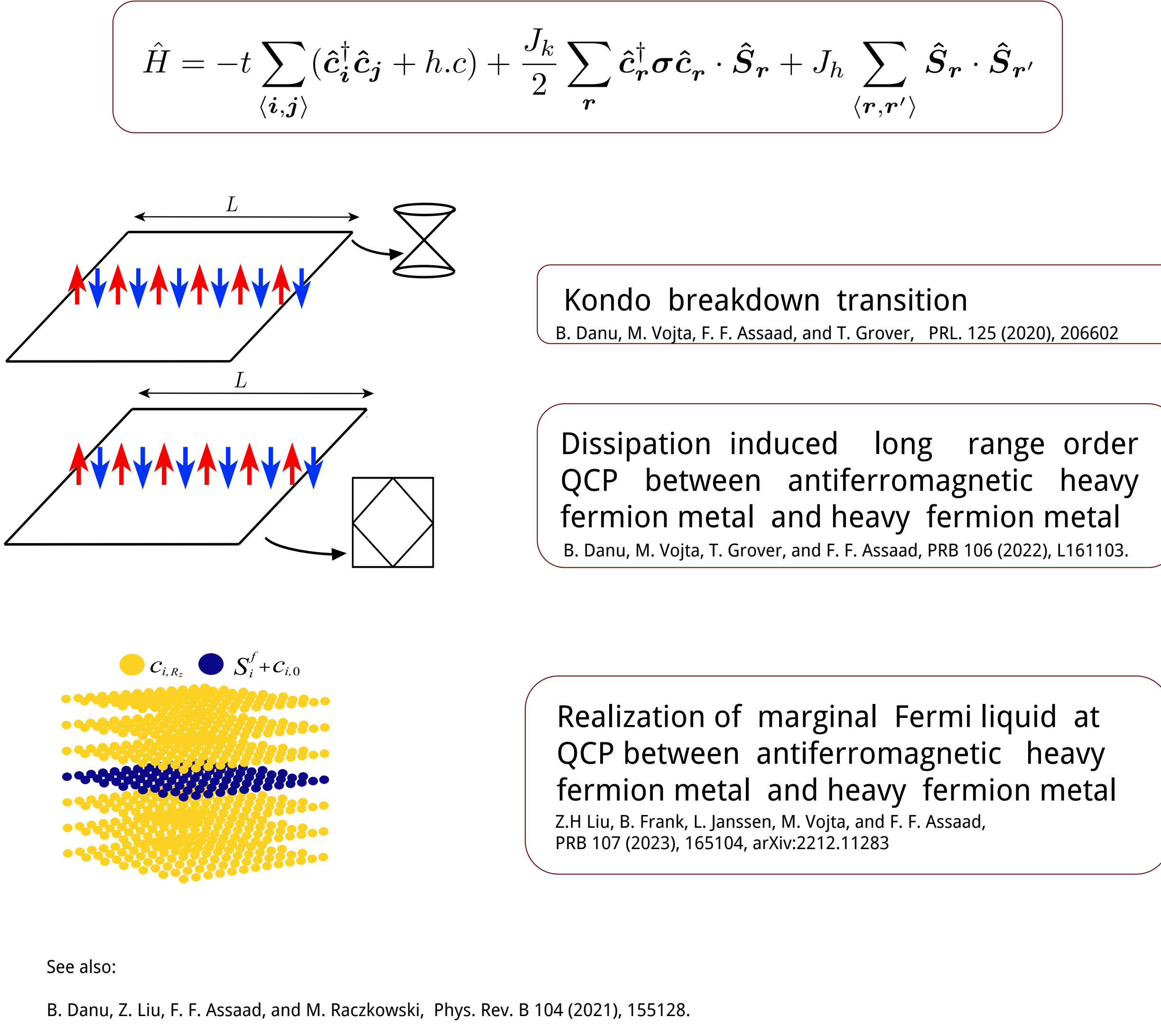
Goals:

- ✓ Generic and flexible package for auxiliary field Monte Carlo simulations of fermion-boson lattice models
- ✓ Efficient implementation on modern HPC systems
- ✓ Benchmarking and golden standards

Dirac systems



SU(N) Kondo-Heisenberg



The ALF package

Kinetic

$$\hat{H} = \sum_{k=1}^{M_T} \sum_{\sigma=1}^{N_{\text{col}}} \sum_{s=1}^{N_{\text{fl}}} \sum_{x,y} \hat{c}_{x\sigma s}^\dagger T_{xy}^{(ks)} \hat{c}_{y\sigma s} + \sum_{k=1}^{M_V} U_k \left\{ \sum_{\sigma=1}^{N_{\text{col}}} \sum_{s=1}^{N_{\text{fl}}} \left[\left(\sum_{x,y} \hat{c}_{x\sigma s}^\dagger V_{xy}^{(ks)} \hat{c}_{y\sigma s} \right) + \alpha_{ks} \right] \right\}^2 + \sum_{k=1}^{M_I} \hat{Z}_k \left(\sum_{\sigma=1}^{N_{\text{col}}} \sum_{s=1}^{N_{\text{fl}}} \sum_{x,y} \hat{c}_{x\sigma s}^\dagger I_{xy}^{(ks)} \hat{c}_{y\sigma s} \right) + \hat{H}_{\text{Ising}}$$

Potential (sum of perfect squares)

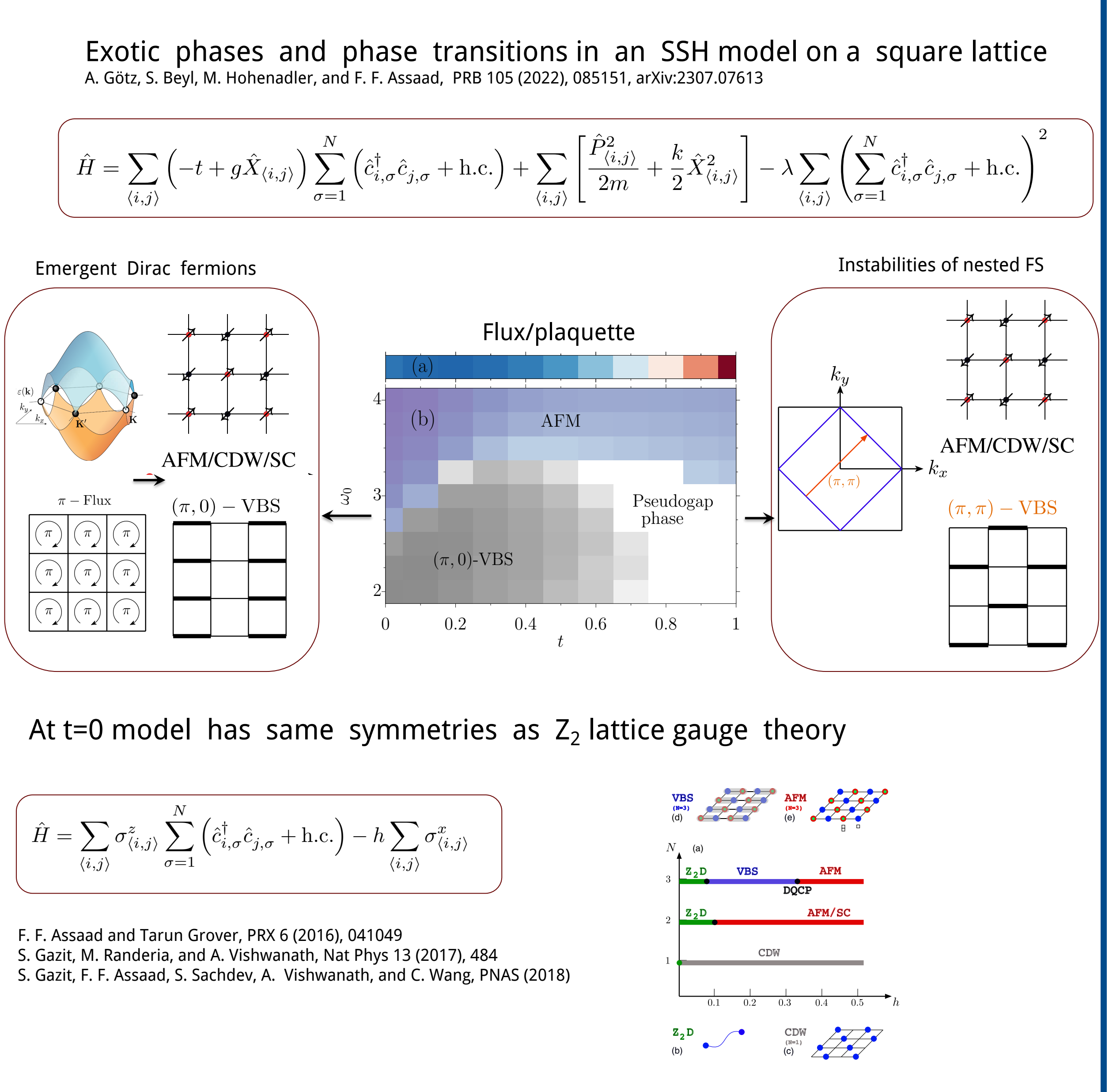
Coupling of fermions to Ising field

- Model can be specified at minimal programming cost
- MPI/OpenMP implementation
- Fortran 2008 standard
- SU(N_{col}) symmetric in colors σ -index
- Arbitrary Bravais lattice for d=1, 2
- Block diagonal in flavors, N_{fl} s-index
- Projective and finite T approaches

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Electron-phonon/Lattice gauge



Frustrated spin systems

